

Joint 3-class Lung Segmentation — DynUNet / 3D U-Net, ex-LIDC (NLST + NSCLC only) — sliding-window (128^3)

1. Task

End-to-end 3-class semantic segmentation of a full chest CT into {background, lung, nodule}. Same task as the paper's `joint_dynunet_ex_lidc` — the only difference is inference recipe: **sliding-window over 128^3 patches at native resolution** rather than a single forward on a 256^3 resample.

- **Input** (1, H, W, D) CT at native voxel grid.
- **Training input** 128^3 patches sampled via `RandCropByLabelClassesd` with class-balanced ratios `[bg, lung, nodule] = [0.2, 0.3, 0.5]`, `num_samples = 4` per volume.
- **Inference** MONAI `sliding_window_inference`, ROI (128, 128, 128), `overlap = 0.5`, `sw_batch_size = 4`, Gaussian blending, output at input resolution.
- **Output** (3, H, W, D) softmax logits: 0 = bg, 1 = lung, 2 = nodule.

2. Model

Property	Value
Architecture family	DynUNet / 3D U-Net
Spatial dims	3
Input channels	1
Output channels	3 (softmax)
6-level auto-configured	strides=[1,2,2,2,2], kernel=3
Trainable parameters	16,538,563

3. Data

Source	Role
NLST	train + val + test
NSCLC-Radiomics	train + val + test
LIDC-IDRI	excluded (no GT lung labels available)

- **Split**: `unified_ex_lidc.json` — patient-grouped, dataset-stratified.
- **Preprocessing**: none beyond intensity clipping / normalization; the full CT enters the sliding-window inference at its native voxel grid.

4. Training

Setting	Value
Loss	Multiclass Focal Tversky + weighted CE ($\alpha=0.3$, $\beta=0.7$, $\gamma=2.0$, $\lambda_{CE}=0.1$, class weights [1, 1, 100])
Optimizer	Adam
Learning rate	1×10^{-5}
Weight decay	1×10^{-5}
LR schedule	Cosine annealing, $T_{max} = 400$, $\eta_{min} = 1 \times 10^{-6}$
Batch size	2 volumes $\times$ 4 patches = 8 patches / step
Patch size	128^3 (class-biased random crop, ratios [bg, lung, nodule] = [0.2, 0.3, 0.5])
Epochs	400
Random seed	42
Mixed precision	bf16
Augmentation	3D flips, 90° rotations, elastic rotation, zoom, intensity scale/shift, Gaussian noise/blur, contrast
Hardware	1 $\times$ NVIDIA H100 80 GB (fat01)

Best checkpoint recorded at **epoch 394 / 400**.

5. Evaluation protocol

Evaluated on the `unified_ex_lidc` **test** split (129 series). All voxel-level metrics micro-averaged over the whole split. Per-class metrics computed against argmax predictions.

6. Results

Ticket-required metrics (3-class average)

Metric	Value
mean IoU (3-cls)	0.7188
mean IoU (lung + nodule)	0.5808
Accuracy	0.994797
Macro Precision (lung + nodule)	0.5963
Macro Recall (lung + nodule)	0.9348

Per-class breakdown

Class	Dice (micro)	Recall	Precision	IoU
background	0.9973	0.9963	0.9984	0.9947
lung	0.9734	0.9729	0.9740	0.9482
nodule	0.3517	0.8968	0.2187	0.2134

Per-case Dice distribution — nodule class (129 test cases)

Statistic	Value
Mean	0.3700
Std	0.2553
Min	0.0000
p25	0.1439
Median	0.3558
p75	0.5903
Max	0.8997

Voxels evaluated: 9,789,871,397. Cases: 129.

7. Notes on interpretation

Lung Dice at ≈ 0.98 is essentially saturated across all architectures and inference recipes — the lung is a large, well-defined foreground class. Nodule Dice is the real signal.

Sliding-window inference at 128^3 trades global lung context for preserved local resolution. On this corpus the trade tends to be neutral-to-negative on nodule Dice compared to the 256^3 resize baseline — patches see only $\sim 15\%$ of a typical lung volume, which limits anatomy-conditional reasoning.

8. Reproducibility

- Config `/home/werner/nodule-training-demo/configs/joint_dynunet_ex_lidc_sw.yaml`
- Checkpoint `stage3_joint/checkpoints_dynunet_ex_lidc_sw/best_model.pth`
- Metrics JSON `/tmp/claude-1002/-home-werner-nodule-segmentation/c2faa0a0-5a75-48f5-8288-377df40092b7/scratchpad/sw_metrics/joint_dynunet_ex_lidc_sw.json`
- Command `python stage3_joint/eval_metrics.py --config /home/werner/nodule-training-demo/configs/joint_dynunet_ex_lidc_sw.yaml --checkpoint stage3_joint/checkpoints_dynunet_ex_lidc_sw/best_model.pth --split test`